# Solution Requirements

| | |
|---|---|
| **Date** | 2 July 2026 |
| **Team Id** | XXXXX |
| **Project Name** | Opti Crop: Smart Agricultural Production Optimization Engine |
| **Maximum marks** | 4 marks |

## Define Functional And non – functional requirements:

- **Functional requirements (FR)** define specific behaviours, functions or features of the systems (what the system should do).
- **Non -functional requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviour (How the system should be).

  Fill out the descriptions and necessary details for each category provided below.

## Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description | Priority |
|---|---|---|---|
| 1 | Authentication | User logs in securely to access the system. | High |
| 2 | Authorization Levels | Farmer can access crop prediction and recommendations. | High |
| 3 | External Interfaces | Connects to soil, weather, and crop datasets. | High |
| 4 | Transaction Processing | Processes user input and generates crop predictions. | High |
| 5 | Reporting | Displays crop recommendations and prediction results. | Medium |

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|---|---|---|---|
| 1 | Performance & Speed | Generate predictions quickly. | Response within **2 seconds** |
| 2 | Scalability | Support multiple users simultaneously. | Up to **500 users** |
| 3 | Security & Data Privacy | Protect user and farm data. | Secure login and encrypted data |
| 4 | Reliability & Availability | Ensure system is available most of the time. | **99% uptime** |
| 5 | Usability & Accessibility | Easy-to-use interface for farmers. | Simple and user-friendly design |
| 6 | Other | Easy to maintain and update. | Supports future enhancements |

| S.No | Requirement Category | Requirement Description | Priority |
|---|---|---|---|
| 6 | Business Rules | Recommends crops based on soil and weather conditions. | High |

## Step 2: Non-Functional Requirements (NFR)